Experimental Demonstration of the Quantum Zeno Effect

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This is my wonderful abstract.

Introduction—The quantum Zeno effect was first theoretically predicted by Misra and Sudarshan in 1977 while investigating probabilities for state decays [1]. The effect predicts that repeated measurements of a quantum state cause repeated wavefunction collapse into the measured state, so it cannot evolve from that state.[1] The effect most often applies to time-dependent state evolution, but the principle can be expanded into time-independent regimes [2].

One such regime is the case of optical polarization, where standard predictions predict no transmission between horizontally and vertically polarized light. However, with the inclusion of optical polarizers, the quantum Zeno effect predict the photon wavefunctions will successively collapse into the polarizer direction. With infinite polarizers, essentially all photons will be transmitted despite the input and output polarization being 90° apart. The result of the quantum Zeno effect is often used to optimize interaction-free measurements [3, 4].

Here,

Theory—The Zeno effect will be demonstrated here by considering the evolution between polarization states. The basis states in this case are $|H\rangle$ and $|V\rangle$. An arbitrary linear polarization state, then, can be written as

$$|\psi\rangle = \cos(\theta)\,|H\rangle + \sin\theta\,|V\rangle\,, \tag{1}$$

where θ is the angle of polarization.

We then aim to calculate the magnitude of the state that transmits through the polarizer. Allowing the photon to interact with the polarizer is effectively "measuring" the state of the photon, which causes its wavefunction to collapse into $|H\rangle$. Thus, in other words, we want to calculate the magnitude of state projected onto the polarizer axis. Without loss of generality, we will assume the polarizer is set to the $|H\rangle$ axis. After all, if the polarizer set at a different angle α, this is equivalent to simply rotating the polarization state by $-\alpha$. The projection operator [5] onto the $|H\rangle$ state is

$$P_H = |H\rangle\langle H|\,. \tag{2}$$

Thus, projecting the arbitrary linear polarization onto $|H\rangle$ gives

$$\begin{aligned} P_H\,|\psi\rangle &= |H\rangle\langle H|\left(\cos(\theta)\,|H\rangle + \sin\theta\,|V\rangle\right) \\ &= \cos(\theta)\,|H\rangle\,. \end{aligned} \tag{3}$$

The magnitude of this state is

$$\langle\psi|\psi\rangle = \cos^2(\theta), \tag{4}$$

which gives the probability that a given photon is transmitted. If there is a stream of photons entering a detector,

$$\langle\psi|\psi\rangle \propto \Gamma, \tag{5}$$

where Γ is the coincidence count rate. Thus, the relation between the input and output count rate is given by

$$\Gamma(\theta) = \Gamma_{\max}\cos^2(\theta). \tag{6}$$

For classical theories of light as an electromagnetic wave, this relation considering $\Gamma \leftrightarrow I$ is also known as Malus's law [6].

In this experiment, we wish to verify this relation is true for single photons as it traverses multiple polarizers, each collapsing the photon's wavefunction into its polarization direction. If there are n polarizers set so that the angle θ between them is consistent, this is equivalent to projecting successively onto different polarization states. Thus, the coincidence count rate for n polarizers with angle difference θ is

$$\Gamma(\theta, n) = \Gamma_{\max}\cos^{2n}(\theta). \tag{7}$$

In the case where the total angle difference is 90°, the relation between n and θ is

$$\theta = \frac{90^\circ}{n} = \frac{\pi}{2n}. \tag{8}$$

This leads to expected expression for the coincidence count rate for the experiment [3],

$$\Gamma(n) = \Gamma_{\max}\cos^{2n}\left(\frac{\pi}{2n}\right), \tag{9}$$

or equivalently,

$$\Gamma(\theta) = \Gamma_{\max}\cos^{\frac{\pi}{\theta}}(\theta). \tag{10}$$

It is critical to note that for increasing n, the rate of coincident photons also increases. Interestingly, for $n \to \infty$, $\Gamma \to \Gamma_{\max}$, meaning that all photons pass through despite the initial and final polarization angles being 90° apart. Thus, the quantum Zeno effect predicts that as more polarizers are added, more of the photons will transmit through, as the more frequent observations force wavefunction collapse in the polarizer bases.

[1] One might refer to the effect as "a watched state never boils."

Experimental Methods—To ensure measurements are characteristic of the quantum Zeno effect, coincident photons must be created and measured. The process of spontaneous parametric down-conversion (SPDC) was used to generate two photons with wavelength 810 nm from a single pump photon with wavelength 405 nm [7–9]. We use beta-barium borate, which causes type I SPDC which ensures the down-converted photons are in the same polarization state [10]. Thus, these two output photons (sent to arm 0 and arm 1) are present in the state

$$|\psi\rangle = |H\rangle_0 |H\rangle_1 . \tag{11}$$

The SPDC efficiency is $\approx 10^{-11}$, so few photons are down-converted. The rest are blocked by a filter and do not continue through the optical setup.

Using coupling mirrors, the photons are redirected towards fiber couplers attached to two inputs on a photon counter, shown in Fig. 1. In arm 0, between the mirror and fiber coupler are four identical polarizers on rotation mounts, which are arbitrarily adjustable and nominally set to the vertical direction parallel to $|V\rangle$. Ensuring the number of physical polarizers present is constant helps to remove the confounding variable of a nonconstant transmission coefficient. The photons in arm 1 are simply coupled into an optical fiber directly as a control.

The photon counter is able to detect the single count rates and coincidence count rates, which is used in order to reduce the effects of background light reaching the coupler. Simultaneous clicks in arm 0 and arm 1 within the coincidence window are recorded as a coincident photon.

Experiments were performed with the quED Entanglement Demonstrator produced by qutools GmbH [8]. The photon counter is the qutools GmbH quCR Control and Read-out Unit [11], provided with the quED apparatus.

To begin the experiment, the laser diode was set to its operating current of 38 mA. Beam alignment was performed to optimize single count rates and coincident count rates by adjusting the coupling mirror and fiber collimator. Once the alignment of the laser was optimized, the quantum Zeno effect experiment could begin. To increase the number of relevant polarizers, the first n polarizers are set at angles according to Equation 8. The count rates were then tabulated and analyzed.

Results—This experiment was performed for up to 4 polarizers, set at equal angles from 0° to 90°. Thus, $\Delta\theta$ ranged from 90° to 22.5°. Error analysis was performed assuming a Poisson distribution for the number of counts,

$$\sigma = \sqrt{N}, \tag{12}$$

and an error of 0.5° in the angles, which were marked in 1° increments on the rotation mount. The accidental count rate was also corrected for using the relation,

$$\Gamma_{\text{accidental}} \approx \tau \Gamma_0 \Gamma_1, \tag{13}$$

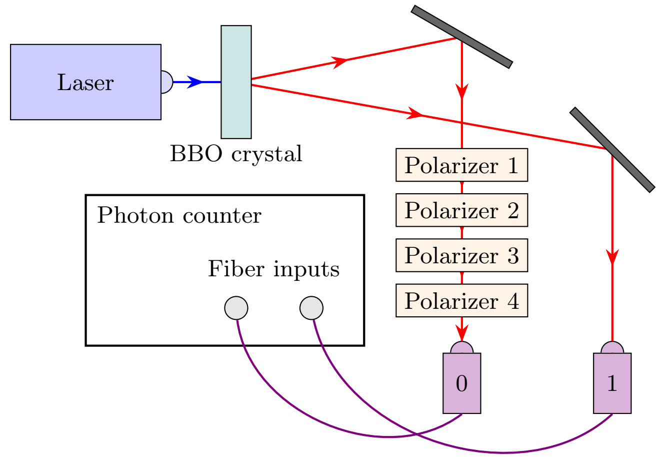

FIG. 1. Experimental setup. The laser diode outputs 405 nm horizontally-polarized light, which passes through a nonlinear down-conversion crystal (beta-barium borate, or BBO), which causes the incident photons to undergo spontaneous parametric down-conversion [10]. This then generates pairs of horizontally-polarized 810 nm photons, which then are separated into two arms. The photon in arm 0 passes through 4 successive polarizers, whereas the photon in arm 1 passes through free space. At the end of each arm, the photons are coupled into optical fibers and led into the input of a photon counter that detects coincident photons.

where τ is the coincidence window (here, 40 ns [8]) and Γ_0 and Γ_1 are the single count rates in arm 0 and 1, respectively. This accidental count rate, along with its calculated error through Poisson statistics, was subtracted for each datum. The coincidence count rate as it relates to the number of polarizers is shown in Fig. 2.

These data were fit to a modification of Equation 9,

$$\Gamma(n) = A \cos^{2n}\left(\frac{\pi}{2n}\right) + C, \tag{14}$$

with A and C free fit parameters. The model showed a fair fit to $A = 1250(34)$ counts/s and $C = 8(4)$ counts/s, with $R^2 = 0.9952$ and reduced $\chi^2 = 4.2447$. For statistical comparison, a linear fit was also performed. The fitted slope was found to be $224(39)\,\mathrm{s}^{-1}$.

For the sake of comparison, the coincidence count rate was also measured for all polarizers set parallel to $|H\rangle$ to simulate the $n \to \infty$ case. This count rate was found to be 1159(11) counts/s in this case. Similarly, the count rate for all polarizers set parallel to $|V\rangle$ was measured to be 14.1(12) counts/s. This gives a measurement of the number of dark counts and counts due to nonideal polarizer transmission.

Conclusion—With a single polarizer, the angle between them was 90° and shows no transmission, as expected. As the number of polarizers increases and the angle difference decreases, we see a monotonically increasing trend characteristic of the expectations of the

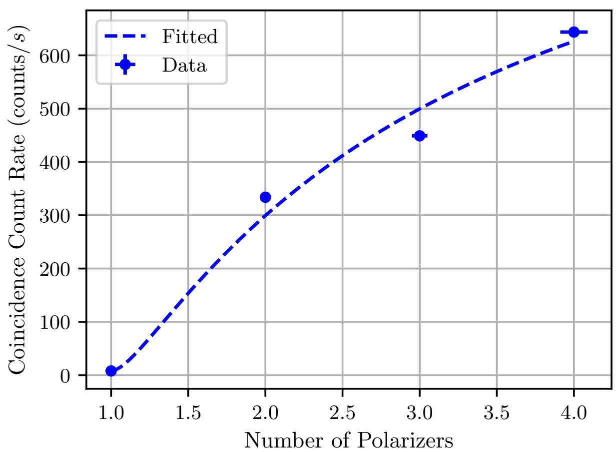

FIG. 2. Coincidence count rate as a function of number of polarizers. Fit performed to Equation 14.

quantum Zeno effect. The quantum Zeno effect-free case predicts a zero slope in case of a linear fit, as increasing n would not increase the number of transmitted photons. Statistical analysis of the slope compared to the null reveals a violation of the null hypothesis by $5.80\,\sigma$. Thus, we observe evidence of the quantum Zeno effect.

The fit to the Malus's law prediction is fairly strong, with low statistical confidence arising due to the relatively low number of data points. Also, the limit of the fit function (Equation 14) for $n \to \infty$ is given by $\Gamma = A + C$, which in this case is 1173(77) counts/s. Comparing to the measured estimated value $n \to \infty$ value of 1159(11) counts/s, we see excellent agreement to $0.18\,\sigma$. The parameter C is approximately equal to the number of dark counts, which agrees to $1.08\,\sigma$. Thus, the number of dark counts is fairly underestimated by the fit model.

Summary—

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